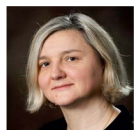




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Editorial

From bark to bytes: artificial intelligence transforming veterinary medicine

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Artificial intelligence is beginning to permeate our daily lives, and veterinary medicine is no exception. Artificial intelligence is a broad umbrella term encompassing the many ways computer algorithms can perform difficult tasks that previously required human intelligence. Veterinary professionals may take different approaches with this new technology, from enthusiastic engagement to skeptical avoidance. This special issue introduces busy veterinary professionals to AI principles and applications in veterinary medicine and research.

On April 19 to 21, 2024, nearly 200 veterinarians, scientists, students, educators, and engineers gathered at Cornell University for the first Symposium on Artificial Intelligence in Veterinary Medicine (SAVY) and shared how AI is already changing companion animal health, livestock health, population medicine, and One Health. We are excited to share some highlights of the ideas presented at SAVY with *AJVR* readers in this supplemental issue.

In roundtable discussions and facilitated workshops, SAVY participants explored the future benefits and challenges that AI will bring to veterinary medicine. Several themes emerged from these group discussions. First, there is a pressing need for education and awareness on AI in veterinary medicine. In many cases, veterinarians are already using AI-powered tools in their practices without fully understanding how these technologies work and the potential impact on diagnostic accuracy, patient and client privacy, and animal health. Artificial intelligence should be added to veterinary curriculums and continuing education so that clinicians are prepared to safely integrate these tools into veterinary medicine. Second, ethical guidelines, regulations, and performance standards for AI in veterinary medicine are lacking. Third, translational research is needed to bridge the gap between AI research and real-world veterinary medicine. Finally, the success of AI applications hinges on access to diverse, extensive, and organized data resources, necessitating the development of data collection and analysis systems.

Authors in this special issue of *AJVR* and of companion reviews in *JAVMA* discuss many of these themes and propose possible paths forward for responsible and innovative integrations of AI into veterinary medicine. Here's a primer on how you can use this special issue to advance your understanding of AI.

Start with a gentle introduction to learn about the areas where AI and machine learning are likely to impact veterinary medicine and some of the scientific limitations and ethical considerations of using these new technologies.¹ Next, learn about how AI may be used to support government regulatory processes and safety surveillance for products used in veterinary medicine.²

Dive into specific use cases, including AI-powered sensors and models for predicting bovine respiratory disease,^{3,4} computer vision for dog-breed identification,⁵ large language model prompts to generate educational messages about antimicrobial resistance,⁶ and machine learning models for imaging interpretation, early disease detection,⁷ and disease risk factor identification⁸ that promise to improve pet health and longevity. Learn from a survey of your peers about benefits and challenges in integrating AI into your veterinary practice.⁹ Farmers and veterinarians can peek into the future, when advances in big data and digitization could significantly improve livestock health and welfare.

Artificial intelligence is also showing up in research processes, potentially making veterinary research more efficient and impactful. For example, scientists are using computer vision and deep learning models to speed up research tasks, generating a deeper understanding of animal behavior.¹⁰ Machine learning prediction models can fill in missing antimicrobial resistance data, enhancing our resistance surveillance systems.¹¹ Finally, AI can be used to predict infectious disease burdens, providing a warning to epidemiologists and veterinarians of future risks to animal and human health.¹²

The papers in this special issue are just the beginning, the "early adopters" of AI. There is still a lot of work to be done to establish AI as a trustworthy

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and effective tool in clinical practice and veterinary research. Whether you embrace AI tools with optimism or a healthy dose of skepticism, this special issue can be your starting point for engaging with this technology that will transform veterinary medicine.

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